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Continuation of 4.1

Proof. The preceding square is a cocartesian diagram, with f open. In particular f is a monomorphism, whence f' is a monomorphism and the square is cartesian as well. The morphism $F' \amalg G \to G'$ is onto. The pull-back of f' by $f' : F' \to G'$ is an isomorphism, since f' is a monomorphism, and hence is open. The pull-back of f' by $G \to G'$ is just f, open by assumption. It follows that f' is open. $\square$

We now fix a class $\mathcal{E}$ of open embeddings $U \to X$ in (QP/G). We require the following stability properties.

Definition Conditions 34. *1. If $U \to U' \to X$ are open embeddings and $U \to X$ is in $\mathcal{E}$, then $U' \to X$ is in $\mathcal{E}$.*

2. If $U \to X$ is an open embedding in $\mathcal{E}$, and if $f : X' \to X$ is étale, inducing an isomorphism from $f^{-1}(Z)$ to Z, where Z is the complement of U in X, then $f^{-1}(U) \to X'$ is in $\mathcal{E}$.

The classes $\mathcal{E}$ that will be used are the following: the open embeddings $U \to X$ with X smooth; the open embeddings $U \to X$ with X smooth and the action of G free on $X - U$, equivalently X is the union of U and the open subset on which the action of G is free; and, when working in (Sch/S), all open embeddings.

Definition 7. *A morphism $f : F \to G$ is $\mathcal{E}$-solid if it is a composite*

$$F = F_0 \longrightarrow F_1 \longrightarrow \cdots \longrightarrow F_n = G$$

where each $F_i \to F_{i+1}$ is deduced by push-out from some $h_U \to h_X$, with $U \subset X$ in $\mathcal{E}$. A sheaf F is solid if $\emptyset \to F$ is $\mathcal{E}$-solid. In the pointed context, a pointed sheaf is pointed $\mathcal{E}$-solid if the morphism $* \to F$ is $\mathcal{E}$-solid.

Example 1. *For U open in X, let h_X/h_U be the sheaf h_X with the subsheaf h_U contracted to the base point. If $U \to X$ is in $\mathcal{E}$, then $* \to h_X/h_U$ is solid: it is the push-out of $h_U \to h_X$ by $h_U \to *$. Thom spaces are of this form: from a vector bundle E on T, one contracts, in the total space of E, the complement of the zero section to a point.*

The class of solid morphisms is the smallest class, closed under compositions and push-outs, which contains all $h_U \to h_X$, $U \subset X$ in $\mathcal{E}$. By Proposition 33, solid morphisms are open.

For a monomorphism $F \to G$ of sheaves, define G/F to be the pointed sheaf obtained by contracting F to a point. Thus one has a cocartesian square

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \\ * & \longrightarrow & G/F. \end{array}$$

By transitivity of push-outs, any cocartesian diagram

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \\ F' & \longrightarrow & G' \end{array}$$

induces an isomorphism $G/F \simeq G'/F'$.

Proposition 35. *A morphism of sheaves $f : F \to G$ is $\mathcal{E}$-solid if and only if it is a composite*

$$F = F_0 \longrightarrow F_1 \longrightarrow \cdots \longrightarrow F_n = G$$

of monomorphisms such that each F_{i+1}/F_i is isomorphic to some h_X/h_U, with $U \subset X$ in $\mathcal{E}$.

Proof. If a morphism $F \to G$ is deduced by push-out from $U \to X$, then G/F is isomorphic to h_X/h_U. This gives the "only if" direction.

Conversely, suppose first that we have a cocartesian square

$$\begin{tikzcd} F \arrow[r] \arrow[d] & G \arrow[d] \\ * \arrow[r] & h_X/h_U \end{tikzcd} \tag{35.1}$$

and that the natural map $h_X \to h_X/h_U$ lifts to G. Then $F \to G$ is obtained by push-out from $h_U \to h_X$. Indeed, (35.1) is cartesian as well as cocartesian, so that we have a cartesian square

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[d] \\ F \arrow[r] & G. \end{tikzcd}$$

If G' is obtained from F by push-out from $h_U \to h_X$,

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[r, equal] \arrow[d] & h_X \arrow[d] \\ F \arrow[r] & G' \arrow[r] & G, \end{tikzcd}$$

then $G'/F \simeq G/F$, and it follows that $G' \simeq G$.

It is enough, more generally, to have an étale map $v : \widetilde{X} \to X$, inducing an isomorphism from $\widetilde{X} - v^{-1}(U)$ to $X - U$, together with a lifting of $h_{\widetilde{X}} \to h_X/h_U$ to G. If $\widetilde{U} = v^{-1}(U)$, then $h_{\widetilde{X}}/h_{\widetilde{U}} \to h_X/h_U$ is an isomorphism. This is checked most directly on the fibre functors defined by G-local henselian schemes: a morphism $T \to X$ which does not map to U lifts uniquely to $\widetilde{X}$. Under these assumptions $F \to G$ is a push-out of $h_{\widetilde{U}} \to h_{\widetilde{X}}$. By the second stability property of $\mathcal{E}$, the open embedding $\widetilde{U} \to \widetilde{X}$ is in $\mathcal{E}$.

We reduce the general case to this one. Suppose $f : F \to G$ is a monomorphism with $G/F \simeq h_X/h_U$, $U \to X$ in $\mathcal{E}$. The cocartesian square

$$\begin{tikzcd} F \arrow[r] \arrow[d] & G \arrow[d] \\ * \arrow[r] & G/F \end{tikzcd} \tag{35.2}$$

gives an epimorphism $* \amalg G \to h_X/h_U$. The natural section of h_X/h_U on X can therefore be lifted locally either to $*$ or to G. Thus, for a filtration

$$\emptyset = C_{n+1} \subset C_n \subset \cdots \subset C_1 \subset C_0 = X$$

by closed equivariant subschemes, there are étale maps $T_i \to X$, with equivariant sections over $C_i - C_{i+1}$, and liftings of $h_{T_i} \to h_X/h_U$ to $*$ or to G. We may start with $X_0 = U$, where the lifting is to $*$, and then shrink the T_i so that the succeeding liftings are to G and induce isomorphisms over the strata.

Since $F \to G$ is a monomorphism, (35.2) is also cartesian. Pulling back the composite

$$* \longrightarrow h_{X_1}/h_U \longrightarrow h_{X_2}/h_U \longrightarrow \cdots \longrightarrow h_X/h_U$$

gives a factorization of $F \to G$ in which each quotient is of the form $h_{T_i}/h_{T_i'}$, and the maps $h_{T_i} \to h_X/h_U$ lift to the intermediate sheaves. Hence each step is a push-out of an open embedding in $\mathcal{E}$, and solidity follows. $\square$

Remark 1. *Equivalently, $F \to G$ is $\mathcal{E}$-solid if and only if the pointed sheaf G/F is an iterated extension of sheaves h_X/h_U, with $U \to X$ in $\mathcal{E}$: there are morphisms*

$$* = E_0 \to E_1 \to \cdots \to E_n = G/F$$

with each E_{i+1}/E_i of the form h_X/h_U.

Proposition 36. *If $f : F \to G$ is open and G is $\mathcal{E}$-solid, then f is $\mathcal{E}$-solid.*

Proof. We say "solid" for $\mathcal{E}$-solid. Let $(*)$ be the property of a sheaf G that every open $F \to G$ is solid. If G is solid, it lies at the end of a chain

$$\emptyset = G_0 \to G_1 \to \cdots \to G_n = G$$

in which each $G_i \to G_{i+1}$ is a push-out of some $h_U \to h_X$, $U \subset X$ in $\mathcal{E}$. We prove by induction on n that G_i satisfies $(*)$.

For $n = 0$, $G_1 = h_X$ is representable and $\emptyset \to X$ is in $\mathcal{E}$. If $F \to h_X$ is open, it is $h_U \to h_X$ for U open in X, hence solid by Condition 34(1).

It remains to check the induction step. In a cocartesian square

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[d] \\ G' \arrow[r] & G \end{tikzcd} \tag{36.1}$$

the map $h_U \to h_X$ is a monomorphism, and the square is therefore cartesian as well as cocartesian. Let $F \to G$ be open. Pull back (36.1) by $F \to G$. We obtain a cartesian and cocartesian square

$$\begin{tikzcd} h_V \arrow[r] \arrow[d] & h_Y \arrow[d] \\ F' \arrow[r] & F \end{tikzcd} \tag{36.2}$$

where Y is open in X and $V = U \cap Y$. The diagram

$$\begin{tikzcd} F' \arrow[r] \arrow[d] & F \arrow[r] \arrow[d] & h_Y/h_V \arrow[d] \\ G' \arrow[r] \arrow[d] & G \arrow[r] \arrow[d] & h_X/h_U \arrow[d] \\ G'/F' \arrow[r] & G/F \arrow[r] & h_X/(h_U \cup h_Y) \end{tikzcd}$$

expresses G/F as an extension of $h_X/h_{U\cup Y}$ by G'/F'. By the induction hypothesis applied to G', and by Remark 1, it follows that $F \to G$ is solid. Here $U \cup Y \to X$ lies in $\mathcal{E}$. $\square$

Proposition 37. *The pull-back of a solid morphism by an open morphism is solid. In particular, if $F \to G$ is open and G is solid, then F is solid.*

Proof. Since the pull-back of an open morphism is open, it suffices to check the assertion for a solid morphism obtained as a push-out of $h_U \subset h_X$:

$$\begin{array}{ccc} h_U & \longrightarrow & h_X \\ \downarrow & & \downarrow \\ G' & \longrightarrow & G. \end{array}$$

After pulling back by an open morphism $F \to G$, we obtain a cocartesian square

$$\begin{array}{ccc} h_{U'} & \longrightarrow & h_{X'} \\ \downarrow & & \downarrow \\ F' & \longrightarrow & F \end{array}$$

with U' open in U and X' open in X. This shows that $F' \to F$ is solid. $\square$

Suppose now that two classes $\mathcal{E}$ and $\mathcal{E}'$ of open embeddings satisfying Conditions 34 are given. Define $\mathcal{E} \times \mathcal{E}'$ to be the smallest class, stable under Conditions 34, containing the open embeddings

$$(U \times X') \cup (X \times U') \subset X \times X',$$

for $U \subset X$ in $\mathcal{E}$ and $U' \subset X'$ in $\mathcal{E}'$.

Example 2. *If $\mathcal{E}$ is the class of all open embeddings and $\mathcal{E}'$ is the class of those $U' \subset X'$ for which G acts freely outside U', then $\mathcal{E} \times \mathcal{E}' = \mathcal{E}'$.*

Proposition 38. *If the pointed sheaves F and F' are respectively $\mathcal{E}$- and $\mathcal{E}'$-solid, then $F \wedge F'$ is $\mathcal{E} \times \mathcal{E}'$-solid.*

Proof. By assumption, F is an iterated extension, in the sense of Remark 1, of pointed sheaves h_{X_i}/h_{U_i}, $U_i \subset X_i$ in $\mathcal{E}$. Similarly F' is built from $h_{X'_j}/h_{U'_j}$, $U'_j \subset X'_j$ in $\mathcal{E}'$. The smash product $F \wedge F'$ is then an iterated extension, for instance in lexicographic order, of

$$(h_{X_i}/h_{U_i}) \wedge (h_{X'_j}/h_{U'_j}) \simeq h_{X_i \times X'_j}/h_{(U_i \times X'_j) \cup (X_i \times U'_j)}.$$

Hence it is $\mathcal{E} \times \mathcal{E}'$-solid. $\square$

Definition 8. *A morphism is called ind-solid relative to $\mathcal{E}$ if it is a filtered colimit of $\mathcal{E}$-solid morphisms.*

Exercise 39. Take G to be the trivial group. A section on T of a push-out

$$\begin{array}{ccc} h_U & \longrightarrow & h_X \\ \downarrow & & \downarrow \\ F & \longrightarrow & G \end{array}$$

can be described as follows. For an open subset V of T and a section s of F on V, consider on the small Nisnevich site T_{Nis} the presheaf $a(V, s)$ which sends $Y \to T$ to the set of morphisms $f : Y \to X$ such that $f^{-1}(U) = Y^{-1}(V)$ and such that the restriction is compatible with s. A section of G on T is given by:

1. an open subset V of T;

2. a section s of F on V;

3. a section of $i_*(a(V,s)_{\mathrm{Nis}})$ on $T-V$, where $i : T-V \to T$ is the closed embedding.

Exercise 40. In the notation of Exercise 39, if F is a sheaf for the étale topology, so is G. For fixed (V,s), the datum in (3) is an étale sheaf. This follows from the usual criterion for a Nisnevich sheaf to be étale: for $x \in T$ and for a finite separable extension $K/k(x)$, after extension of the residue field from the henselization $\mathcal{O}^h_{T,x}$, the corresponding functor of étale neighbourhoods must be an étale sheaf on $\operatorname{Spec} k(x)$.

Exercise 41. It follows from Exercises 39 and 40 that if $f : F \to G$ is ind-solid, and if F is étale, then G is étale. In particular, solid sheaves, as well as pointed solid sheaves, are étale sheaves.

Remark 2. *The same formalism of open and solid morphisms holds in the site of all schemes of finite type over S, with the étale topology.*

4.2 A criterion for preservation of local equivalences

We work with pointed sheaves on QP/G. The aim of this section is to prove the following result.

Theorem 6. *Let a be a functor from pointed sheaves to pointed sets. Suppose that a commutes with all colimits, and that for every open embedding $U \to X$, the map*

$$a((h_U)_+) \longrightarrow a((h_X)_+)$$

is a monomorphism. If $f : F_\bullet \to G_\bullet$ is a local equivalence and the terms F_n and G_n are pointed ind-solid, then $a(f)$ is a weak equivalence.

Suppose that

$$Q = \begin{array}{ccc} B & \longrightarrow & Y \\ \big\downarrow & & \big\downarrow \\ A & \longrightarrow & X \end{array}$$

is an upper distinguished square. Adding a base point, we obtain Q_+. The morphism $K_Q \to X_+$ is a local equivalence. Let us check that $a(K_Q) \to a(X_+)$ is a weak equivalence. Since a commutes with coproducts, this morphism is obtained from the commutative square

$$\begin{array}{ccc} a((h_B)_+) & \longrightarrow & a((h_Y)_+) \\ \big\downarrow & & \big\downarrow \\ a((h_A)_+) & \longrightarrow & a((h_X)_+) \end{array}$$

by applying the same construction as in (23.3). The square is cocartesian because Q is. The top horizontal map is a monomorphism, hence the square is homotopy cocartesian, and the claim follows. Since a commutes with colimits, this special case implies the following.

Lemma 15. *If f is in the $\bar{\Delta}$-closure of the morphisms $(K_Q)_+ \to X_+$ as above, then $a(f)$ is a weak equivalence.*

Proposition 42. *Proof of Theorem 6.*

Proof. For any pointed sheaf F, the morphism $S(F) \to F$ is a local equivalence: for every connected $X \in QP/G$, $S(F)(X) \to F(X)$ is a weak equivalence by Construction 29. Hence, for a local equivalence $f : F \to G$, the square

$$\begin{array}{ccc} S(F) & \longrightarrow & S(G) \\ \downarrow & & \downarrow \\ F & \longrightarrow & G \end{array}$$

is a commutative square of local equivalences. By Lemmas 15 and 12, $a(S(f))$ is a weak equivalence. It remains to prove that $a(S(F)) \to a(F)$ is a weak equivalence, and similarly for G. Since a and S commute with filtered colimits, ind-solid reduces to solid, and by the inductive definition of solidity it suffices to prove the following lemma. □

Lemma 16. *Let $U \to X$ be an open embedding. Suppose that in a cartesian square of pointed sheaves*

$$\begin{array}{ccc} (h_U)_+ & \longrightarrow & (h_X)_+ \\ \downarrow & & \downarrow \\ F & \longrightarrow & G \end{array}$$

the sheaf F is such that $a(S(F)) \to a(F)$ is a weak equivalence. Then the same holds for G.

Proof. Consider the cocartesian square

$$\begin{array}{ccc} S((h_U)_+) & \longrightarrow & S((h_X)_+) \\ \downarrow & & \downarrow \\ S(F) & \longrightarrow & S(G). \end{array}$$

The top morphism is a monomorphism. It follows that this square is pointwise homotopy cocartesian, and $S \to S(G)$ is a local equivalence. The functor i^*i_* of Construction 29 sends a monomorphism to a coprojection of the form $F \to F \amalg \coprod (h_{U_\alpha})_+$. Consequently each S_n is of the form $(\coprod h_{U_\alpha})_+$, and by Lemmas 15 and 12 the map $a(S) \to a(S(G))$ is a weak equivalence. It remains to show that $a(S) \to a(G)$ is a weak equivalence.

Apply a to the morphism from the square just displayed to the original cartesian square. By Construction 29, both

$$S((h_U)_+) \to (h_U)_+, \qquad S((h_X)_+) \to (h_X)_+$$

are homotopy equivalences and remain so after applying a. We assumed $a(S(F)) \to a(F)$ to be a weak equivalence. Since a sends both squares to cocartesian squares with top morphisms monomorphisms, it follows that $a(S) \to a(G)$ is a weak equivalence. Hence so is $a(S(G)) \to a(G)$. □

5 Two functors

5.1 The functor $X \mapsto X/G$

There is a natural morphism of sites

$$p : (QP/G)_{\mathrm{Nis}} \longrightarrow (QP)_{\mathrm{Nis}}$$

given by the functor

$$p^* : \; QP \longrightarrow QP/G, \qquad X \longmapsto (X \text{ with the trivial } G\text{-action}).$$

Indeed p^* commutes with finite limits and sends covering families to covering families. In particular it is continuous: if F is a sheaf on $(QP/G)_{\mathrm{Nis}}$, the presheaf

$$X \longmapsto F(X \text{ with trivial } G\text{-action})$$

is a sheaf on $(QP)_{\mathrm{Nis}}$. The functor p^* has a left adjoint

$$q: \; QP/G \longrightarrow QP, \qquad X \longmapsto X/G.$$

As p^* is continuous, the functor q is cocontinuous, and the inverse-image functor p^* from sheaves on $(QP)_{\mathrm{Nis}}$ to sheaves on $(QP/G)_{\mathrm{Nis}}$ is

$$F \longmapsto (\text{sheaf associated to } X \longmapsto F(X/G)).$$

Proposition 43. *The cocontinuous functor $q : X \mapsto X/G$ is also continuous. Equivalently, if F is a sheaf on QP, then the presheaf $X \mapsto F(X/G)$ on QP/G is a sheaf.*

Proof. By Lemma 2 it is enough to test the sheaf property of $X \mapsto F(X/G)$ for a covering of X obtained by pull-back from a Nisnevich covering $U_i \to X/G$. Passage to quotients commutes with flat base change. Taking X/G as base, this identifies $X \times_{X/G} U_i \to U_i$ with the quotient, by G, equal to U_i. The same holds for multiple intersections. Thus the sheaf property for the covering of X by the $X \times_{X/G} U_i$ is reduced to the sheaf property of F for the covering (U_i) of X/G. □

The functor $q : X \mapsto X/G$ gives rise to a pair of adjoint functors between the categories of presheaves on QP/G and QP. Since q is continuous, it induces a similar pair on sheaves. We write this pair as

$$(p_\#, p^*),$$

so that one has a sequence of adjunctions $(p_\#, p^*, p_*)$. If F on QP/G is representable, $F = h_X$, then $p_\# F = h_{X/G}$. In particular, $p_\#$ sends the final sheaf, or point, on QP/G to the final sheaf on QP, and $(p_\#, p^*)$ is an adjoint pair for pointed sheaves as well. It is clear that $p_\#$ sends solid sheaves to solid sheaves. We also have the following.

Proposition 44. *Let F be a pointed sheaf which is solid with respect to open embeddings $U \subset X$ of smooth schemes such that the action of G on X is free outside U. Then $p_\#(F)$ is solid with respect to open embeddings of smooth schemes.*

Proof. If X' is the open subset of X where the action of G is free, then $U \cup X' \to X$ is in the same class, and, putting $U' = U \cap X'$, a push-out of $U \to X$ is also a push-out of $U' \to X'$. Thus we may suppose that the action is free everywhere. Since $p_\#$ is a left adjoint, it preserves colimits and in particular push-outs. It sends h_U to $h_{U/G}$. Hence the assertion is reduced to the following lemma. □

Lemma 17. *If G acts freely on U, smooth over S, then U/G is smooth.*

Proof. If G is finite étale, for instance S_n, the case most relevant here, this is clear, since $U \to U/G$ is étale. In general, the assumption that G acts freely on U implies that U is a G-torsor over U/G. In particular $U \to U/G$ is faithfully flat. Since U is flat over S, this forces U/G to be flat over S. To check smoothness of U/G over S, it suffices to check it geometric fibre by geometric fibre. For a geometric point $\bar{s}$ of S, smoothness of $(U/G)_{\bar{s}}$ amounts to regularity. Since $U_{\bar{s}}$ is smooth over $\bar{s}$, hence regular, and $U_{\bar{s}} \to (U/G)_{\bar{s}}$ is faithfully flat, this is [4], an application of Serre's cohomological criterion for regularity. □

Proposition 45. *The functor $p_\#$ respects local, respectively $\mathbb{A}^1$-, equivalences between termwise ind-solid simplicial sheaves.*

Proof. Let $f : F \to F'$ be a local equivalence between termwise ind-solid simplicial sheaves on QP/G. To verify that $p_\#(f)$ is a local equivalence, it is enough to check that for every $X \in QP$ and $x \in X$ the map

$$p_\#(F)(\operatorname{Spec} \mathcal{O}^h_{X,x}) \longrightarrow p_\#(F')(\operatorname{Spec} \mathcal{O}^h_{X,x})$$

is a weak equivalence of simplicial sets. Since $p_\#$ is a left adjoint, the functor

$$F \longmapsto p_\#(F)(\operatorname{Spec} \mathcal{O}^h_{X,x}) \tag{45.1}$$

commutes with colimits. For an open embedding $U \to X$ in QP/G, the map $U/G \to X/G$ is again an open embedding, and Theorem 6 applies to (45.1).

Let $f : F \to F'$ be an $\mathbb{A}^1$-equivalence. Consider the square

$$\begin{array}{ccc} S(F) & \longrightarrow & S(F') \\ \downarrow & & \downarrow \\ F & \longrightarrow & F'. \end{array}$$

By the first part of the proposition, $p_\#$ takes the vertical arrows to local equivalences. Since $p_\#$ commutes with colimits, Lemma 13 implies that $p_\#(S(f))$ is in the $\bar{\Delta}$-closure of the class containing the $p_\#((K_Q)_+ \to X_+)$, for Q upper distinguished, and the $p_\#((X \times \mathbb{A}^1)_+ \to X_+)$, for $X \in QP/G$. By Theorem 4 it remains to prove that morphisms of these two types are $\mathbb{A}^1$-equivalences. For the first type this follows from the first half of the proposition. For the second type, it follows from the fact that

$$p_\#((X \times \mathbb{A}^1)_+) \longrightarrow p_\#(X_+)$$

and

$$p_\#(X_+) \times \mathbb{A}^1 \longrightarrow p_\#((X \times \mathbb{A}^1)_+)$$

are mutually inverse $\mathbb{A}^1$-homotopy equivalences. □

Define

$$Lp_\# : \mathcal{H}_{A^1}(QP/G) \longrightarrow \mathcal{H}_{A^1}(QP)$$

(and similarly in the pointed setting) by

$$Lp_\#(F) := p_\#(S(F)),$$

where $S(F)$ is Construction 29. Proposition 45 shows that $Lp_\#$ is well-defined, and that, for a termwise ind-solid F, one has $Lp_\#(F) \simeq p_\#(F)$.

5.2 The functor $X \mapsto X^T$

As in Sect. 3.1, we fix G and S. We also fix $T \in QP/G$, finite and flat over S.

For a presheaf F on QP/G, define F^T to be the internal Hom object $\underline{\mathrm{Hom}}(h_T, F)$. Its value on U is

$$F^T(U) = F(U \times_S T).$$

If F is a sheaf, so is F^T.

Example 6. *Take G and T deduced from the finite group S_n acting on $\{1, \dots, n\}$ by permutations. If F is represented by X, with trivial S_n-action, then F^T is represented by X^n, on which S_n acts by permutation of the factors.*

Remark 1. *If F is representable, and represented by X smooth over S, then F^T is representable, and represented by a smooth G-scheme. More precisely, if F is represented by $X \in QP/G$, consider the contravariant functor on Sch/S of morphisms of schemes from T to X,*

$$U \longmapsto \mathrm{Hom}_S(T \times_S U,\ X \times_S U).$$

This functor is represented by a scheme Y, quasi-projective over S and smooth if X is smooth. The scheme Y carries the evident action of G, and (Y, G) represents F^T. Indeed, by attaching to a morphism $T \to X$ its graph, one maps the functor into the Hilbert functor of finite subschemes of $T \times_S X$ of the same rank as T. This Hilbert functor is represented by a quasi-projective Hilbert scheme. The condition that a finite flat subscheme be the graph of a morphism from T to X is an open condition; it gives the required open subscheme Y. If X is smooth, smoothness of Y follows from the infinitesimal lifting criterion. The G-action is the evident one. For $Z \in QP/G$ one has the chain of identifications

$$\mathrm{Hom}_{QP/G}(Z, Y) \simeq \mathrm{Hom}_S(Z, \mathrm{Hom}(T, X)) \simeq \mathrm{Hom}_S(Z \times_S T, X) \simeq F^T(Z).$$

Let $\mathcal{E}$ be a class of open embeddings in $(QP/G)_{\mathrm{Nis}}$. We will simply say "solid" for $\mathcal{E}$-solid.

Theorem 7. *If F is a solid sheaf on $(QP/G)_{\mathrm{Nis}}$, then F^T is solid.*

If $A \to F$ is open, that is, relatively representable by open equivariant embeddings, there is a natural sequence of sheaves between A^T and F^T. In the situation of Example 6, and for $h_U \to h_X$, these are represented by the open equivariant subschemes of X^n consisting of the n-tuples $(x_1, \ldots, x_n)$ for which at least i of the x_j lie in U.

Formally, a section of F^T over Z is a section s of F over $T \times_S Z$. Let $U(s)$ be the equivariant open subscheme of $T \times_S Z$ on which s lies in A. The sheaf $(F, A)^T_i$ is the subsheaf of F^T consisting of those s such that all fibres of $U(s) \to Z$ have degree at least i. The condition that the degree be at least i is open; hence the inclusion $(F, A)^T_i \subset F^T$ is open. For $i = 0$, this sheaf is F^T; for i large, it is A^T.

Lemma 18. *Suppose that $A \to F$ is deduced by push-out from an open map $B \to C$, so that*

$$\begin{array}{ccc} B & \longrightarrow & C \\ \downarrow & & \downarrow \\ A & \longrightarrow & F \end{array} \tag{45.2}$$

is cocartesian. Then, for each i, the cartesian square

$$\begin{array}{ccc} \textit{fiber product} & \longrightarrow & (A \amalg C, A)^T_i \\ \downarrow & & \downarrow \\ (F, A)^T_{i+1} & \longrightarrow & (F, A)^T_i \end{array} \tag{45.3}$$

is cocartesian as well.

Proof. The site $(QP/G)_{\mathrm{Nis}}$ has enough points. As a consequence of Lemma 2, for each $X \in QP/G$ and each $x \in X/G$, the functor

$$F \longmapsto \varinjlim F(X \times_{X/G} U)$$

where U ranges over the Nisnevich neighbourhoods of x in X/G, is a point. Such points are represented by G-local henselian schemes Z, finite disjoint unions of local henselian schemes essentially of finite type over S, with Z/G local.

We show that (45.3) becomes cocartesian after applying any such fibre functor $F \mapsto F(Z)$. Fix $s \in (F,A)^T_i(Z)$, viewed as a section of F over $T \times_S Z$, and let $U \subset T \times_S Z$ be the open subset on which s lies in A. The assumption means that the degree of the fibre U_z is at least i, independently of the chosen closed point z of Z. It remains to show that, if s is not in $(F,A)^T_{i+1}(Z)$, so that this degree is exactly i, then s has a unique lifting to $(A \amalg C, A)^T_i(Z)$.

The scheme $T \times_S Z$ is a disjoint union of G-local henselian schemes. On each component on which s does not lie in A, cocartesianness of (45.2) gives a unique lift to C. On the components on which s lies in A, the lift is the given section in A. This produces a section of $A \amalg C$ over $T \times_S Z$ lifting s, and it is unique by the same componentwise argument. $\square$

Proof of Theorem 7. Since F is solid, it lies at the end of a sequence

$$\emptyset = F_0 \to F_1 \to \cdots \to F_n = F$$

where each $F_i \to F_{i+1}$ is a push-out of an open embedding in QP/G. We prove by induction on i that for every Z, $(F_i \amalg h_Z)^T$ is solid. For $i = 0$, F_0 is representable, and so is $(F_0 \amalg h_Z)^T$ by the representability statement above. For the induction step, one applies the following lemma to $F_i \amalg h_Z \to F_{i+1} \amalg h_Z$.

Lemma 19. *Let*

$$\begin{tikzcd} h_U \arrow[r] \arrow[d] & h_X \arrow[d] \\ F \arrow[r] & G \end{tikzcd}$$

be a cocartesian square with U open in X. Suppose that, for every Z, $(F \amalg h_Z)^T$ is solid. Then $F^T \to G^T$ is solid.

Proof. Since $F \to G$ is open by Proposition 33, the $(G,F)^T_i$ are defined. It suffices to prove that, for each i, the open morphism

$$(G,F)^T_{i+1} \longrightarrow (G,F)^T_i$$

is solid. By Lemma 18, this morphism sits in a cartesian and cocartesian square

$$\begin{tikzcd} \text{fiber product} \arrow[r] \arrow[d] & (F \amalg h_X, F)^T_i \arrow[d] \\ (G,F)^T_{i+1} \arrow[r] & (G,F)^T_i. \end{tikzcd} \tag{45.4}$$

By assumption $(F \amalg h_X)^T$ is solid; hence $(F \amalg h_X, F)^T_i$ is solid as well, by Proposition 37 applied to the open morphism into $(F \amalg h_X)^T$. The right vertical arrow in (45.4) is open, hence solid by Proposition 36, and the lower horizontal arrow is solid as the push-out of a solid map. $\square$

Example 7. *It is not always true that, if $f : A \to F$ is a solid morphism, then f^T is solid. Take G trivial and T two points, so that $F^T = F^2$. For every sheaf F, the inclusion $F \to F \amalg *$ is solid. Applying $(-)^T$, one obtains*

$$F^2 \longrightarrow F^2 \amalg F \amalg F \amalg *.$$

Pulling back by the natural map from F to one of the two summands F, which is an open map, we see that if f^T were solid, then F would be solid.

Corollary 46. *If $f : F \to G$ is open and G is solid, then*

$$f^T : F^T \longrightarrow G^T$$

is solid. In particular, if G is pointed solid, then G^T is pointed solid.

Proof. The morphism f^T is open, and one applies Theorem 7 and Proposition 36. $\square$

We now define, for pointed sheaves on $(QP/G)_{\mathrm{Nis}}$, a smash variant of the construction $F \mapsto F^T$. If the marked point $* \to F$ is open, it is

$$F^{\wedge T} := F^T/(F,*)^T_1,$$

that is, F^T with $(F,*)^T_1$ contracted to the new base point. This definition can be repeated for any pointed sheaf if, for every monomorphism $A \to F$, $(F,A)^T_1$ is defined as the following subsheaf of F^T: a section s of $F^T(X) = F(X \times T)$ is in $(F,A)^T_1$ if, for every non-empty $X' \to X$, there is a commutative diagram

$$\begin{array}{ccc} X'' & \longrightarrow & X \times T \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

with X'' non-empty and s lying in A on X''.

Example 8. *Under the assumptions of Example 6, if U is open in X with complement Z, and if $F = h_X/h_U$, then $F^{\wedge T}$ is $h_{X^n}/h_{X^n - Z^n}$, where X^n has its natural S_n-action. In particular, if F is the Thom space of a vector bundle E on Y, then $F^{\wedge T}$ is the Thom space of the S_n-equivariant vector bundle $\bigoplus_1^n E$ on Y^n.*

Proposition 47. *If a pointed sheaf F is pointed solid, that is, if $* \to F$ is solid, then $F^{\wedge T}$ is pointed solid.*

Proof. Since F^T is solid, the open morphism $(F,*)^T_1 \to F^T$ is solid by Proposition 36, and $* \to F^{\wedge T}$ is obtained from it by push-out. $\square$

The definition of $F^{\wedge T}$ immediately implies the following.

Lemma 20. *Let Z be a G-local henselian scheme. Then*

$$F^{\wedge T}(Z) = \bigwedge_i F((T \times Z)_i),$$

where the $(T \times Z)_i$ are the G-local henselian components of $T \times Z$.

Proposition 48. *The functor $F \mapsto F^{\wedge T}$ respects local and $\mathbb{A}^1$-equivalences.*

Proof. Let $f : F \to G$ be a local equivalence. To check that $F^{\wedge T} \to G^{\wedge T}$ is a local equivalence, it is enough to test on any G-local henselian Z. By Lemma 20, the map becomes the smash product of the maps

$$F((T \times Z)_i) \longrightarrow G((T \times Z)_i),$$

and hence is a weak equivalence of simplicial sets.

Let f be an $\mathbb{A}^1$-equivalence. By the first part it is enough to show that

$$S(f)^{\wedge T} : S(F)^{\wedge T} \longrightarrow S(G)^{\wedge T}$$

is an $\mathbb{A}^1$-equivalence. We use the characterization of $\mathbb{A}^1$-equivalences in Theorem 5. Since $F \mapsto F^{\wedge T}$ commutes with filtered colimits and preserves local equivalences, it suffices to check that it takes $\mathbb{A}^1$-homotopy equivalences to $\mathbb{A}^1$-homotopy equivalences. This follows from the natural map

$$F^{\wedge T} \wedge (\mathbb{A}^1)_+ \longrightarrow (F \wedge (\mathbb{A}^1)_+)^{\wedge T}.$$

$\square$

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